

Task 1:

```
def calculator(operand1, operand2, operator):  
    if operator == '+':  
        return operand1 + operand2  
    elif operator == '-':  
        return operand1 - operand2  
    elif operator == '*':  
        return operand1 * operand2  
    elif operator == '/':  
        if operand2 != 0:  
            return operand1 / operand2  
        else:  
            return "Error: Division by zero"  
    else:  
        return "Error: Invalid operator"  
result = calculator(10, 5, '+')  
print(result)
```

>>>Output: 15

Task 2:

```
def mask_credit_card(card_number):  
    # Ensure the card number is a string and has at least 4 digits  
    card_number = str(card_number)  
  
    # Mask all but the last four digits  
    if len(card_number) >= 4:  
        masked_number = '*' * (len(card_number) - 4) +  
card_number[-4:]
```

```
else:  
    return "Invalid card number"
```

```
return masked_number
```

Example usage:

```
credit_card = "1234567890123456"  
masked = mask_credit_card(credit_card)  
print(masked)
```

>>>Output: *****3456

Task 3:

```
def apply_discount(original_price, discount_percentage):  
    # Ensure the original price is positive and discount is between  
    0 and 100
```

```
    if original_price < 0 or not (0 <= discount_percentage <=  
    100):
```

```
        return "Invalid input"
```

```
    # Calculate the discount amount
```

```
    discount_amount = (discount_percentage / 100) *  
original_price
```

```
    # Subtract the discount amount from the original price
```

```
    final_price = original_price - discount_amount
```

```
    return final_price
```

```
# Example usage:  
original_price = 100  
discount_percentage = 20  
final_price = apply_discount(original_price,  
discount_percentage)  
print(final_price)
```

>>> Output: 80.0

Task 4:

```
# Initialize an empty dictionary to store expenses
```

```
expenses = {}
```

```
def add_expense():
```

```
    # Input details for the expense
```

```
    date = input("Enter the date (YYYY-MM-DD): ")
```

```
    category = input("Enter the category (e.g., food, travel, entertainment): ").lower()
```

```
    amount = float(input("Enter the amount: "))
```

```
    # If the category already exists, add the amount to the existing list
```

```
    if category in expenses:
```

```
        expenses[category].append((date, amount))
```

```
    else:
```

```
        expenses[category] = [(date, amount)]
```

```
    print("Expense added successfully.\n")
```

```
def display_total_expenses():
```

```
    category = input("Enter the category to view total expenses (e.g., food, travel, entertainment): ").lower()
```

```
    if category in expenses:
        total = sum(amount for date, amount in
expenses[category])
        print(f"Total expenses for {category}: ${total:.2f}\n")
    else:
        print(f"No expenses found for the category
'{category}'.\n")
```

```
def main():
    while True:
        print("Expense Tracker")
        print("1. Add an expense")
        print("2. View total expenses for a category")
        print("3. Exit")

        choice = input("Choose an option (1/2/3): ")

        if choice == "1":
            add_expense()
        elif choice == "2":
```

```
    display_total_expenses()
elif choice == "3":
    print("Exiting Expense Tracker. Goodbye!")
    break
else:
    print("Invalid choice. Please try again.\n")
```

```
# Run the expense tracker
```

```
main()
```